GRADE 2 · BLOCK CODING

Predict Both

Guess where both land, then run to check

Name: .... Date: ....

★ **I can predict where both sprites land when the green flag starts them at the same time.**

Predict first, then run

Good coders PREDICT what their code will do before they run it. Here Bit (green) and Pixel (purple) both start on the green flag ▶ at the SAME time. Read both scripts and PREDICT where each one lands. Then run them to check your prediction.

The number path

0 Bit	1	2	3	4	5 Pixel

Bit

Bit's script

▶ start at 0

forward 2

forward 1

Pixel

Pixel's script

▶ start at 5

back 2

back 1

1 Predict — before you run

Without moving anything yet, PREDICT where each sprite will land. Bit will land on: ____ Pixel will land on: ____

2 Run and check

Now run both scripts from the green flag. Where did each really land? Bit: ____ Pixel: ____. Did your prediction match?

3 Do they meet?

Look at both landing numbers. Do Bit and Pixel land on the SAME number? Circle YES or NO.

Big idea

Predicting both sprites means tracing TWO scripts in your head before running. When the green flag starts them together, each sprite still follows its own script — so a good prediction names both landing numbers.

TEACHER GUIDE · GRADE 2 · BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Students PREDICT where both characters will land before running, then run to check. Predicting both landings means tracing two sets of moves in your head at once. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

How to lead it (about 20 minutes)

1. Show them (you do). Read both sets of moves aloud without moving and say your prediction for each.
2. Do one together. Exercise 1 — predict Bit on 3 and Pixel on 2.
3. Let them try. Students run both to check (Ex 2), then decide if they meet (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 (predict) — Bit: 0, forward 2 → 2, forward 1 → 3 (predict 3). Pixel: 5, back 2 → 3, back 1 → 2 (predict 2).

Exercise 2 (run) — the run confirms Bit on 3 and Pixel on 2.

Exercise 3 — 3 and 2 are different → NO, they do not meet (they land right next to each other).

If students get stuck — and ways to adjust

Watch for: predicting only one character. A full prediction names BOTH landings.

Make it easier: predict Bit, then predict Pixel, one at a time.

Make it harder: ask them to predict whether they meet before running.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

You'll know it worked when

The child predicts both landings, runs to check, and decides whether they meet.